

# Istiaqur Rahman Fuad

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## SUMMARY

Computer science undergraduate and aspiring software and AI engineer, comfortable across the full stack and with large language models. Builds end-to-end web applications and AI features, from fine-tuned models and retrieval systems to LLM-powered tools. Enjoys shipping real products and open-sourcing the work.

## TECHNICAL SKILLS

**Languages:** C, C++, Java, JavaScript, TypeScript, Python, SQL

**Technologies & Tools:** Next.js, FastAPI, tRPC, Prisma, PostgreSQL, Git, Docker

**AI / ML:** PyTorch, Hugging Face, LoRA, LLMs, LangChain, LangSmith, Vector Databases

## EXPERIENCE

### Advanced Machine Intelligence Research (AMIR) Lab

Feb 2026 – Present

*Research Assistant Intern*

*Remote*

- Curated and published **bangla-english-banglish-pairs**, a 2.4M-pair trilingual contrastive dataset (Bangla / Banglish [romanized Bengali] / English) on Hugging Face for low-resource NLP research.
- Trained and open-sourced **TriBne-e5-small**, a trilingual sentence-embedding model for retrieval over code-mixed Bengali text.

### Devcore Technologies

Feb 2025 – Apr 2025

*Next.js Developer Intern*

*Remote*

- Built a modular admin dashboard with end-to-end type-safe APIs (Next.js, tRPC, Prisma, TypeScript) for an e-commerce platform with a 1,000+ product catalog, delivering full product and order CRUD.
- Developed responsive, server-rendered storefront pages with Next.js App Router, improving cross-device usability.

## EDUCATION

### University of Rajshahi

Jan 2022 – Expected 2026

*B.Sc. in Computer Science and Engineering*

*CGPA: 3.58/4.00*

- Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Software Engineering, Artificial Intelligence, Machine Learning

## PROJECTS

### LegalBuddy | [Live](#) | [GitHub](#)

- Tools used:** Python, FastAPI, Next.js, Qdrant, sentence-transformers, Gemini / Groq, Docker
- Built a retrieval-augmented generation (RAG) system over Bangladesh legal acts: a FastAPI backend retrieves grounded statute sections from Qdrant and generates cited answers with Gemini / Groq in a Next.js chat UI.

### PetSoft | [Live](#) | [GitHub](#)

- Tools used:** Next.js, Prisma, TailwindCSS, Stripe, React Hook Form
- Developed a subscription-based pet daycare management system with Next.js and Prisma, featuring secure authentication, Stripe-powered subscription billing, and a responsive interface built with TailwindCSS.

## RESEARCH & PUBLICATIONS

### A Trilingual Contrastive Corpus and Parameter-Efficient Recipe for Romanized Bengali Retrieval

- First author;** under review at *Language Resources and Evaluation* (Springer Nature), 2026. [Demo](#) | [GitHub](#)
- LoRA fine-tuned multilingual-e5-small (1% parameter update, MNRL loss) on a self-built 2.4M-pair trilingual (Bangla / Banglish / English) contrastive corpus for Banglish and cross-script retrieval.
- Reached state-of-the-art on Banglish-spelling and cross-script retrieval, lifting cross-script MRR@10 from 0.50 to 0.98 over the base model and beating BGE-M3, LaBSE, and Embedding Gemma.

## COURSES AND CERTIFICATIONS

- Machine Learning Specialization – Stanford University | DeepLearning.AI (Jul 2023). [certificate](#)
- Deep Learning Specialization – Stanford University | DeepLearning.AI (Oct 2023). [certificate](#)

## ACHIEVEMENTS AND ACTIVITIES

- Award of Merit** – International Blockchain Olympiad (Oct 2024). [certificate](#)
- SDG-9 Silver Medal** – International Blockchain Olympiad (Oct 2022). [certificate](#)